TASK 6: Associate Public IP Address to a window Virtual Machine

Objective

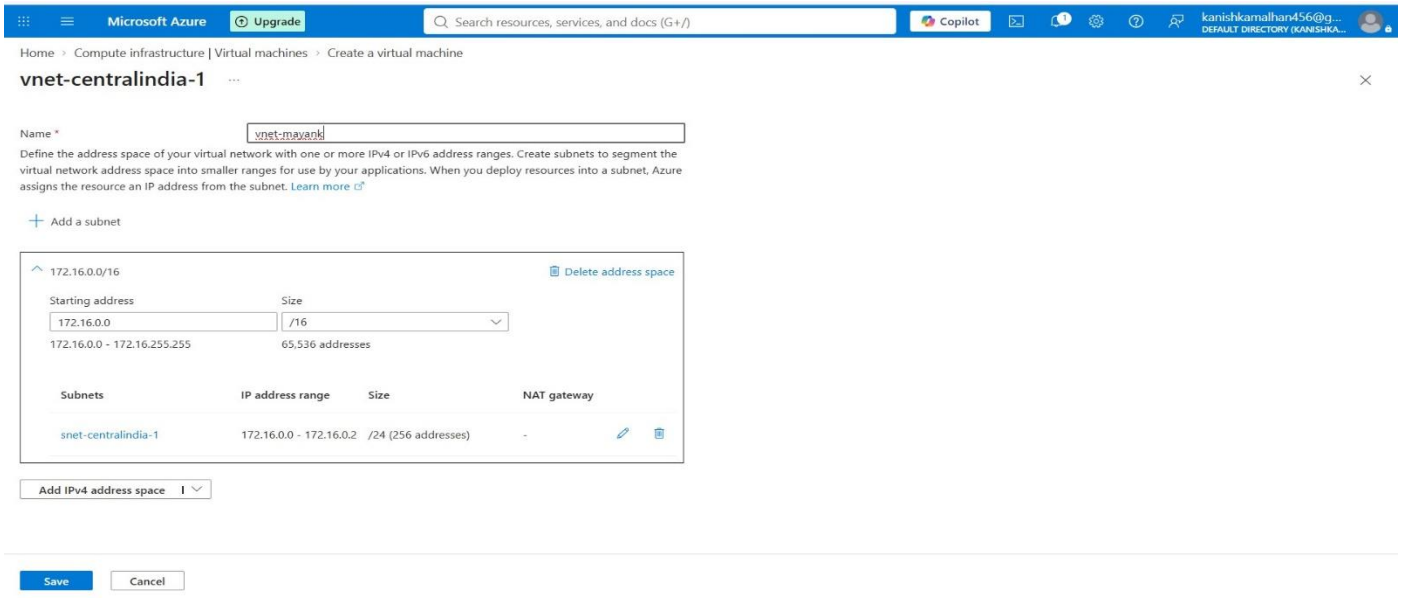
To create and associate a Public IP Address with an Azure Virtual Machine so that the virtual machine can be accessed from the internet using Remote Desktop Protocol (RDP).

Public IP Theory

A Public IP Address is an Azure networking resource that enables communication between Azure resources and the internet. When a Public IP is associated with a Virtual Machine's Network Interface (NIC), users can remotely access the VM from outside the Azure Virtual Network.

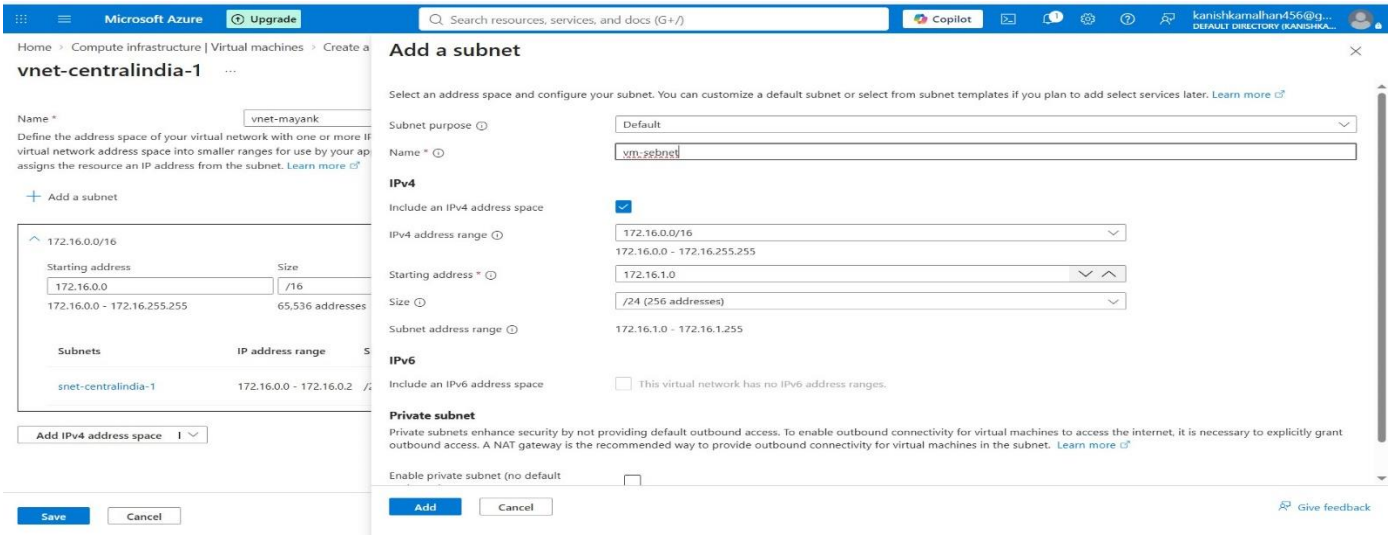
Step 1: Open Network Configuration

Navigate to the Virtual Machine networking configuration and review the network settings associated with the VM.



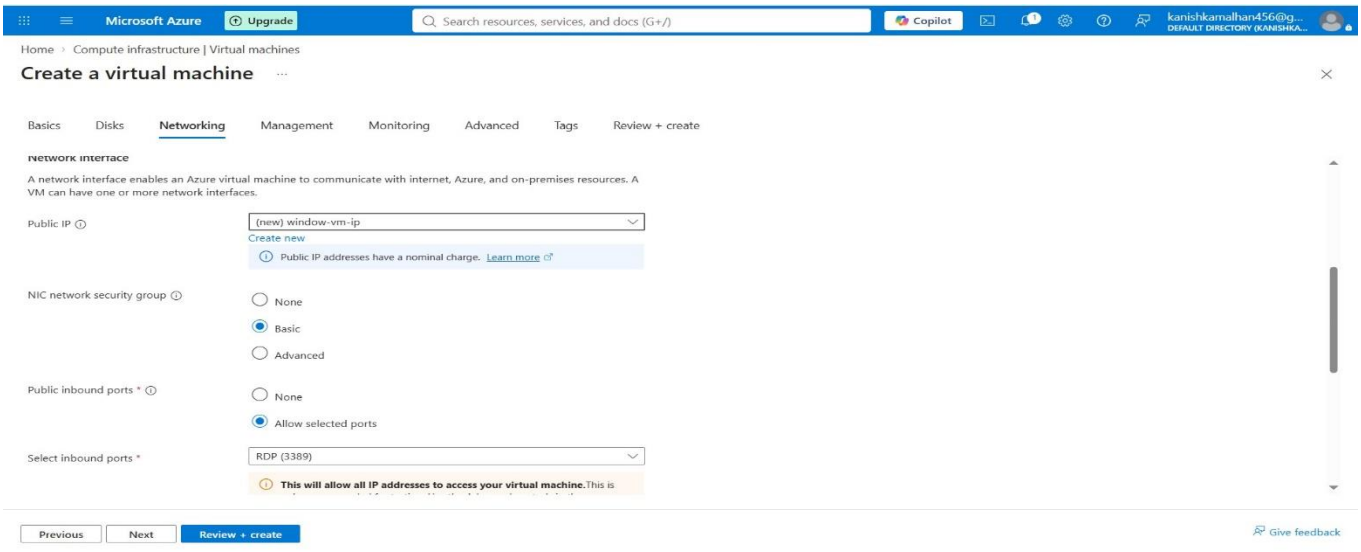
Step 2: Verify Subnet Configuration

Review the subnet in which the virtual machine is deployed and verify network connectivity settings.



Step 3: Create and Associate Public IP Address

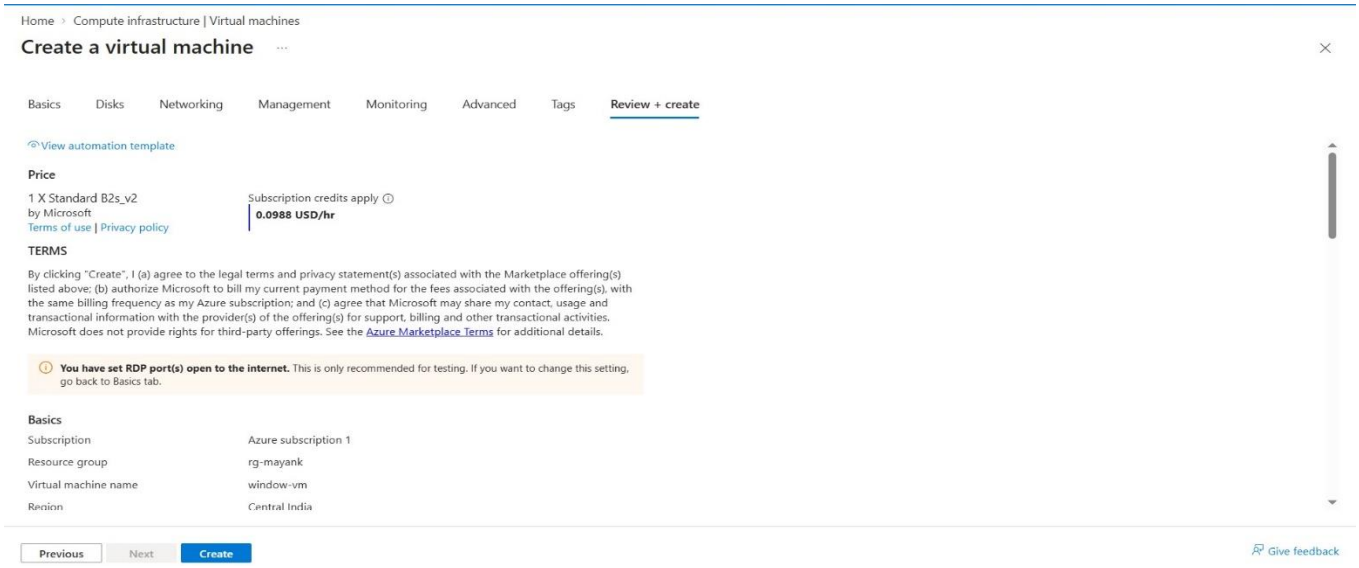
1. In the Networking section, click **Create New Public IP**.
2. Enter the Public IP Name.
3. Associate the Public IP with the Virtual Machine NIC.
4. Configure Network Security Group settings.
5. Allow required inbound traffic such as RDP (3389).



Result: A Public IP Address is successfully created and associated with the Virtual Machine.

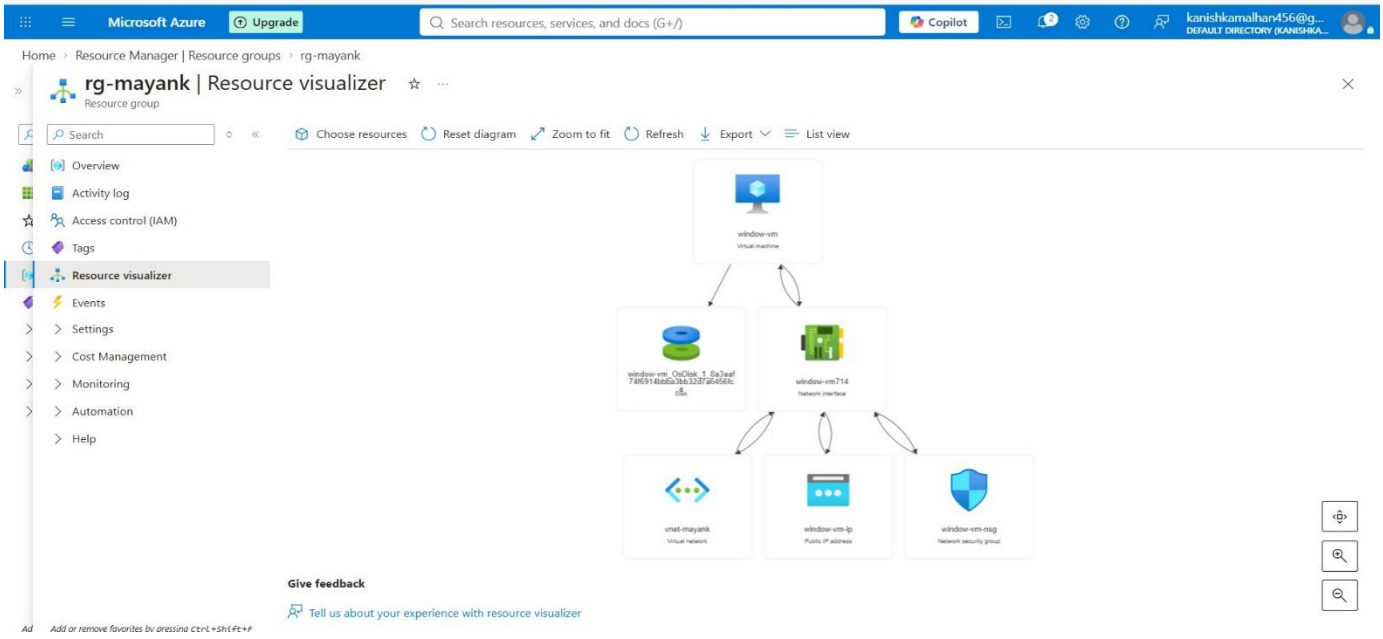
Step 4: Review Configuration

Review the networking configuration and validate that the Public IP resource is attached to the virtual machine.



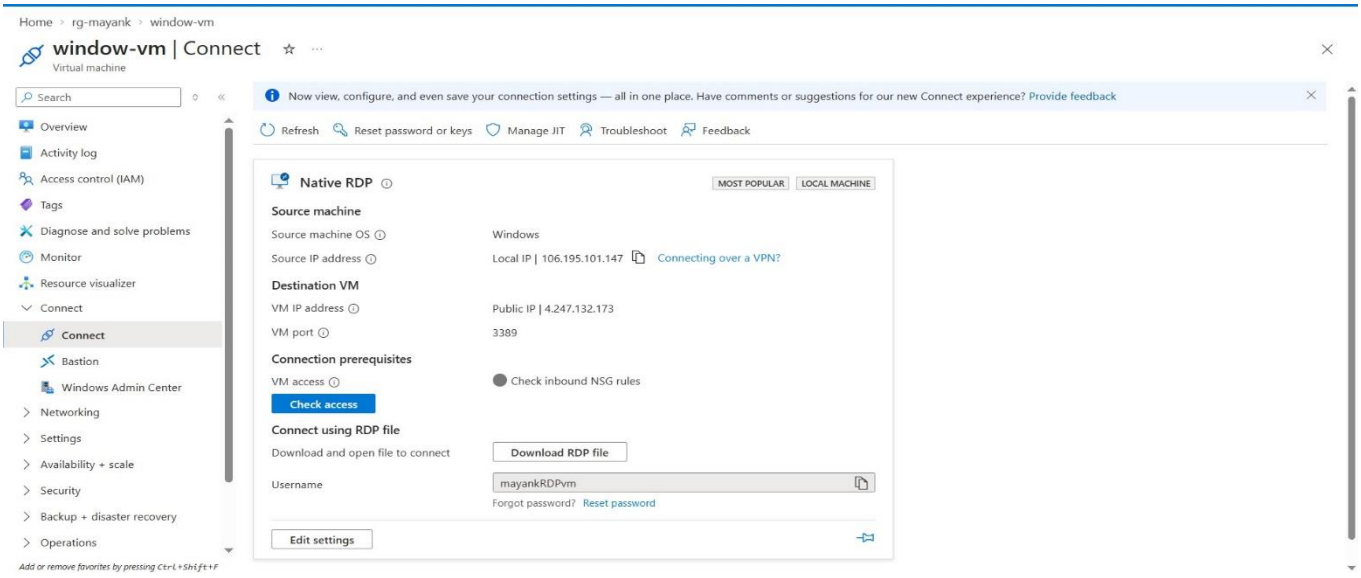
Step 5: Validate Deployment

Verify that the Public IP resource has been successfully deployed and linked with the Virtual Machine.



Step 6: Verify Public IP Connectivity

Navigate to the VM Connect page and verify that the Public IP Address is visible and available for remote connectivity.



Validation

- Public IP Created Successfully
- Public IP Associated with VM
- NSG Configured Successfully
- Public IP Visible in VM Configuration
- Remote Connectivity Available

Conclusion

A Public IP Address was successfully created and associated with the Azure Virtual Machine. The Public IP enables external connectivity to the virtual machine and allows remote administration using services such as Remote Desktop Protocol (RDP). The association was verified successfully through the VM networking and connection settings.

Associate Public IP Address to a Linux Virtual Machine

Objective

To create and associate a Public IP Address with a Linux Virtual Machine in Microsoft Azure to enable remote access using SSH.

Public IP Address Theory

A Public IP Address is an Azure networking resource that enables communication between Azure resources and the internet. By associating a Public IP Address with a Linux Virtual Machine, administrators can securely connect to the VM using SSH over Port 22 for remote management and administration.

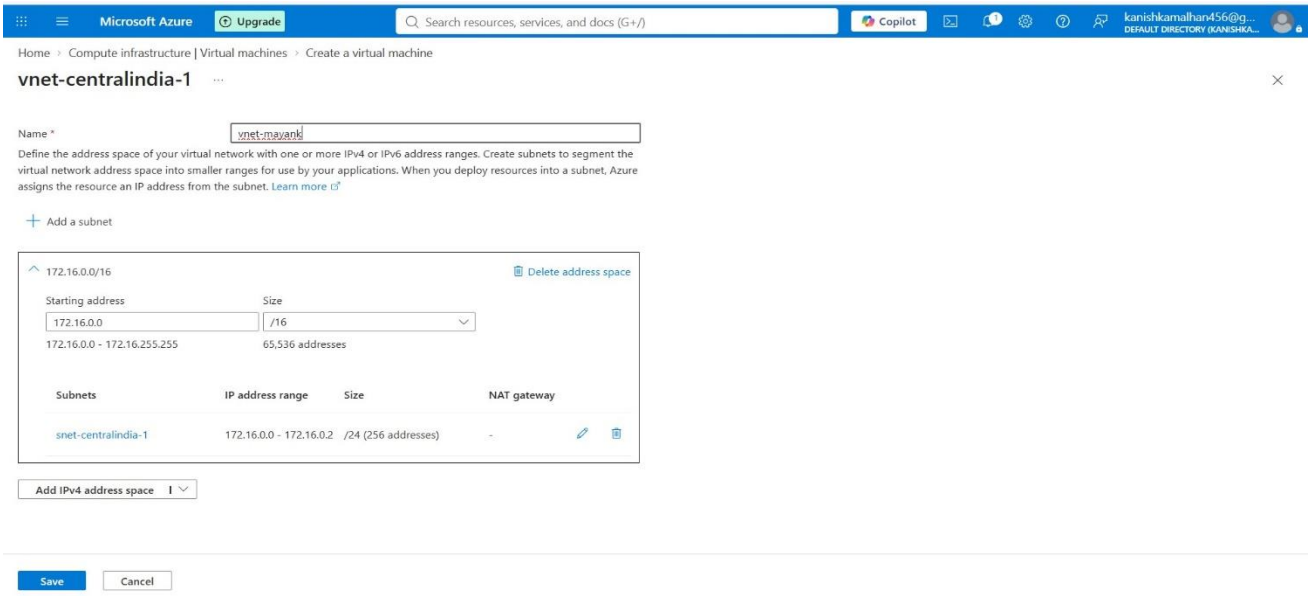
Components Used

- **Linux Virtual Machine (VM):** Provides compute resources.
- **Public IP Address:** Allows external internet connectivity.
- **Network Interface (NIC):** Connects the VM to the network.
- **Virtual Network (VNet):** Enables network communication.
- **Subnet:** Logical segmentation within a VNet.
- **Network Security Group (NSG):** Controls inbound and outbound traffic.
- **SSH (Port 22):** Used for remote administration of Linux Virtual Machines.

Procedure

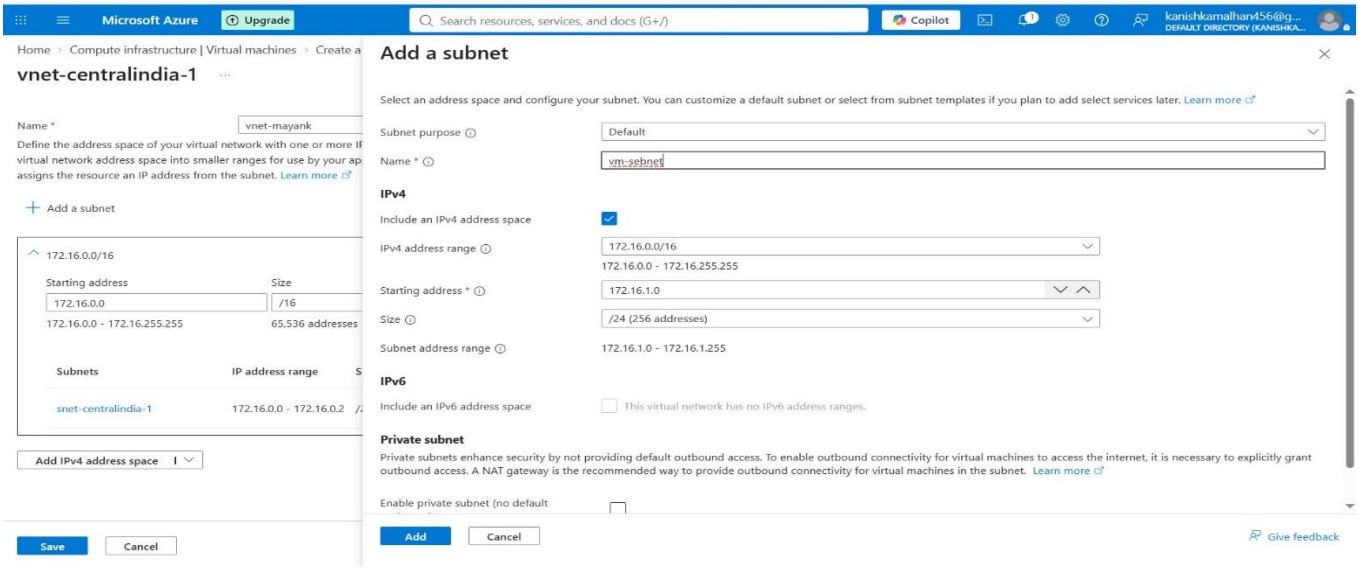
Step 1: Configure Virtual Network

Create or select a Virtual Network for the Linux Virtual Machine.



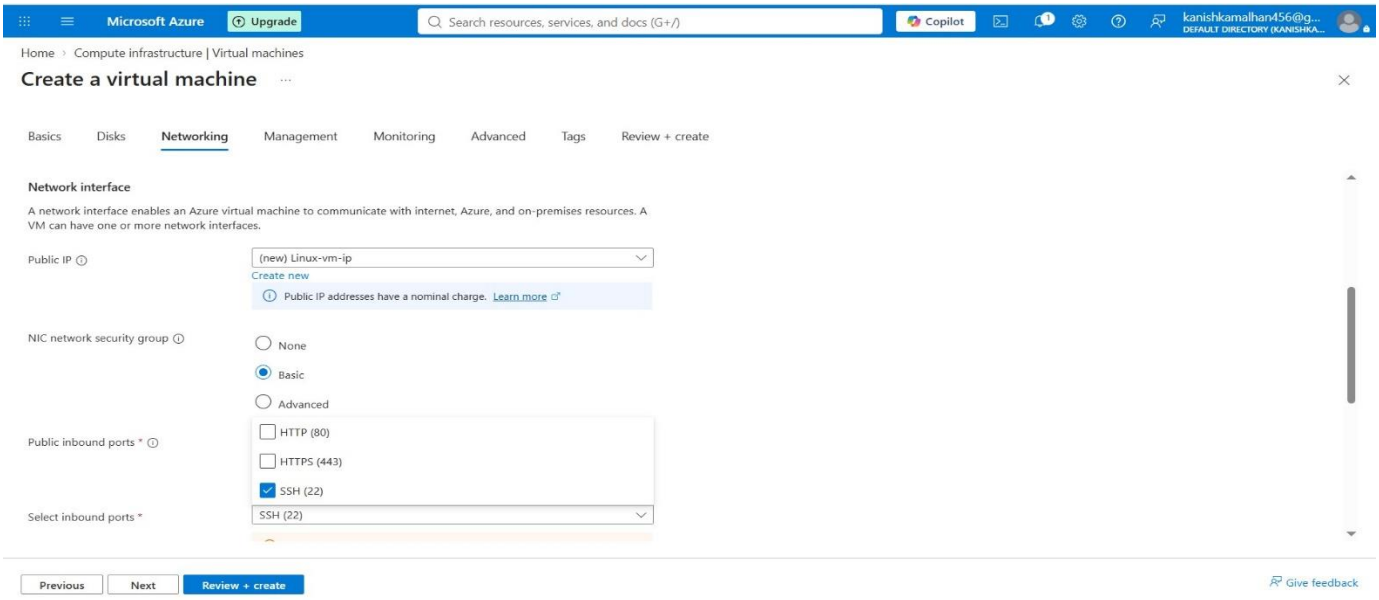
Step 2: Configure Subnet

Create or select a subnet within the Virtual Network.



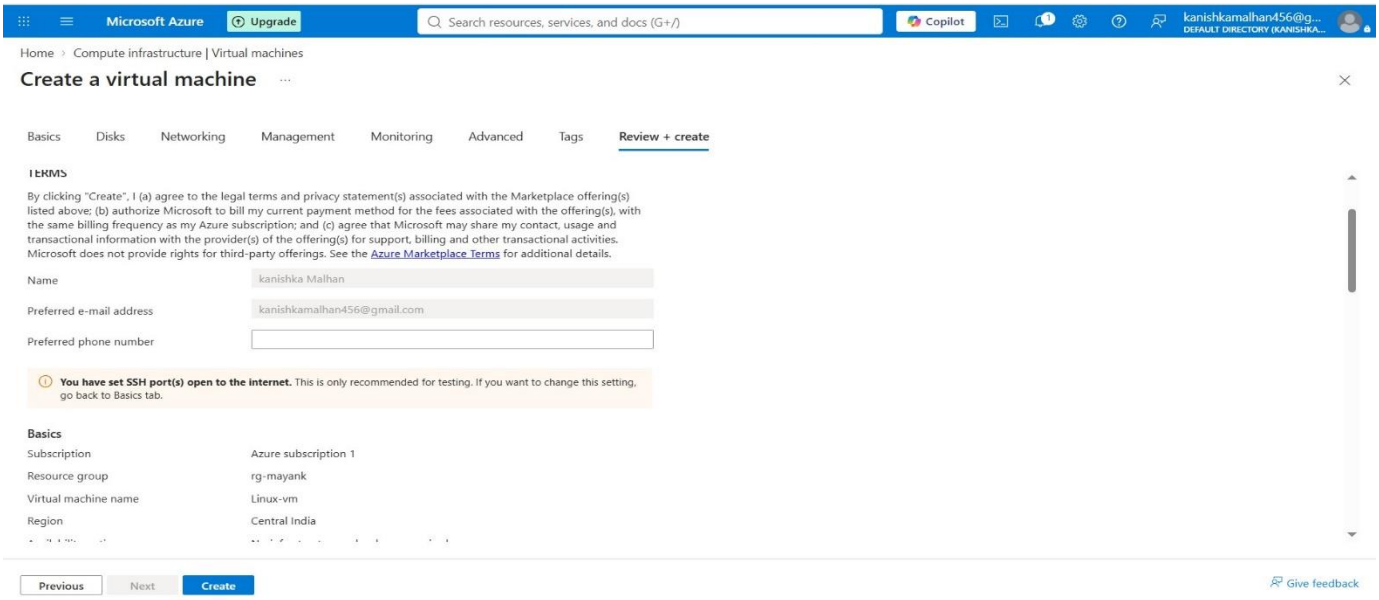
Step 3: Create and Associate Public IP Address

Create a new Public IP Address and associate it with the Linux Virtual Machine Network Interface. Configure Network Security Group settings and allow inbound SSH traffic on Port 22.



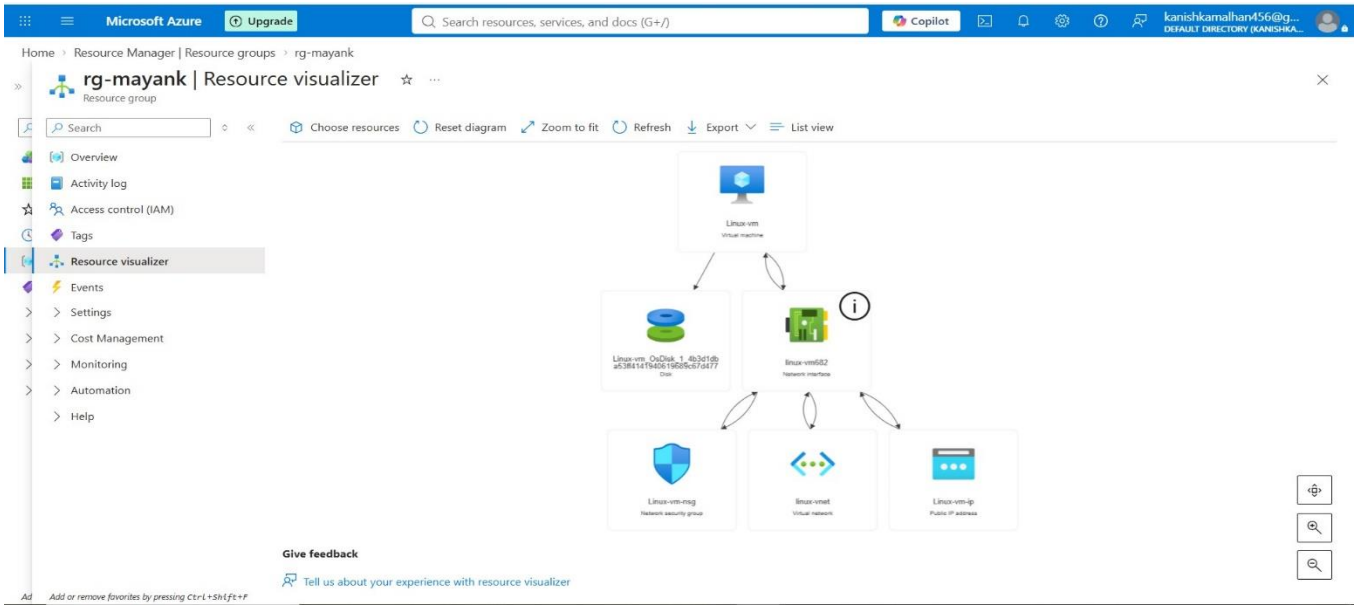
Step 4: Review Configuration

Review all networking settings and validate the Public IP configuration.



Step 5: Verify Resource Deployment

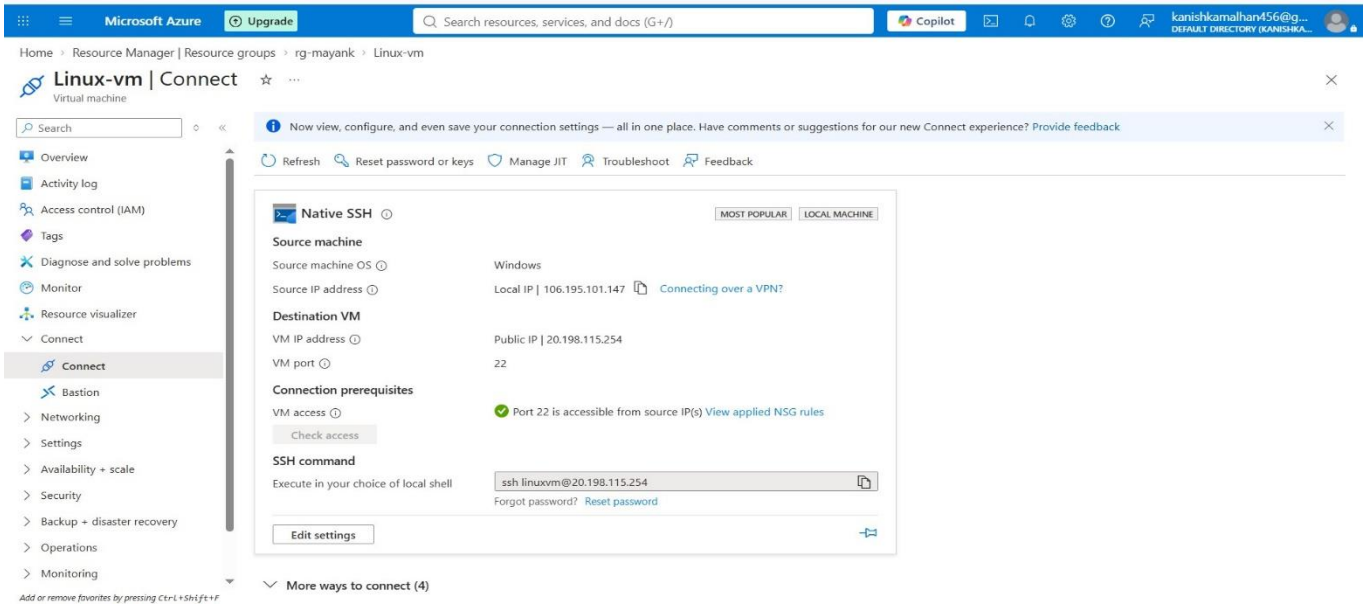
Verify that the Public IP resource has been successfully created and associated with the Linux Virtual Machine.



Step 6: Verify Public IP and SSH Connectivity

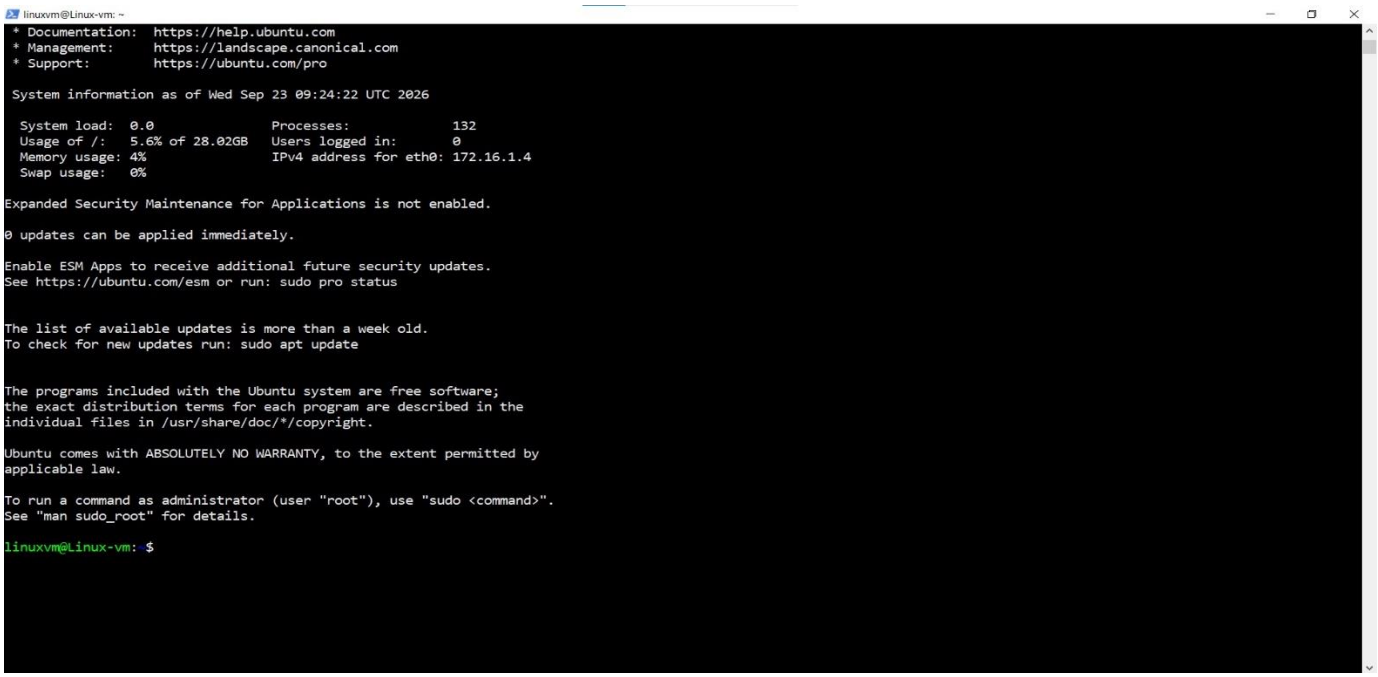
Open the VM Connect page and verify:

- Public IP Address
- SSH Port (22)
- SSH Connection Command



Step 7: Validate Remote Connectivity

Use the SSH command to connect to the Linux Virtual Machine using the assigned Public IP Address.



Validation

- Public IP Address Created Successfully
- Public IP Associated with Linux VM
- Network Security Group Configured
- SSH Port 22 Enabled
- Public IP Visible in VM Configuration
- SSH Connectivity Successful

Conclusion

A Public IP Address was successfully created and associated with the Linux Virtual Machine. The Public IP enabled external connectivity, allowing secure remote administration through SSH over Port 22. The configuration was verified successfully through the VM networking and connection settings.